Prop Let $(F, +, \cdot, <)$ be an ordered field. Then:

① $a > 0 \iff -a < 0$

② If $a, b, c \in F$ satisfy $a < b$ and $c < 0$, then $a \cdot c > b \cdot c$.

③ If $a \in F \setminus \{0\}$ then $a^2 = a \cdot a > 0$.
In particular, $1 > 0$.

④ If $a, b \in F$ satisfy $0 < a < b$, then $0 < b^{-1} < a^{-1}$.

Pf: ① Last time.

② Suppose $a < b$ and $c < 0$. Then

$$c < 0 \overset{①}{\Rightarrow} \left.\begin{array}{l} -c > 0 \\ a < b \end{array}\right\} \overset{O2}{\Rightarrow} \underset{\substack{\| \\ -(a \cdot c)}}{a \cdot (-c)} < \underset{\substack{\| \\ -(b \cdot c)}}{b \cdot (-c)} \quad \text{by lec 6}$$

$$\overset{O1}{\Rightarrow} -ac + (ac + bc) < -bc + (ac + bc)$$

$$\overset{A3, A2}{\Rightarrow} (-ac + ac) + bc < -bc + (bc + ac)$$

$$\overset{A5, A3}{\Rightarrow} 0 + bc < (-bc + bc) + ac$$

$$\overset{A4, A5}{\Rightarrow} bc < 0 + ac$$

$$\overset{A4}{\Rightarrow} bc < ac$$

③ As $a \neq 0$, there are 2 cases (by trichotomy):

- $a > 0 \overset{O2}{\Rightarrow} a \cdot a > 0 \cdot a \overset{\text{lec 6}}{\Rightarrow} a^2 > 0$
- $a < 0 \overset{②}{\Rightarrow} a \cdot a > 0 \cdot a \overset{\text{lec 6}}{\Rightarrow} a^2 > 0$

④ Claim: If $a>0$ then $a^{-1}>0$. Suppose not. Then $\exists\, a>0$ s.t. $a^{-1}\leq 0$ (by trichotomy). Note $a^{-1}\neq 0$ since $a\cdot a^{-1}=1\neq 0$, so $a^{-1}<0$. Now,

$$\left.\begin{array}{l} a>0 \\ a^{-1}<0 \end{array}\right\} \overset{②}{\Rightarrow} a\cdot a^{-1}<0 \overset{M5}{\Rightarrow} 1<0.$$

This contradicts ③. Therefore $a^{-1}>0$.

Next, suppose $0<a<b$. Then

$$\begin{aligned}
0<a<b &\overset{O2}{\Rightarrow} 0\cdot(a^{-1}\cdot b^{-1}) < a\cdot(a^{-1}\cdot b^{-1}) < b\cdot(a^{-1}\cdot b^{-1}) \\
&\overset{M3,\,M2}{\Rightarrow} 0 < (a\cdot a^{-1})\cdot b^{-1} < b\cdot(b^{-1}\cdot a^{-1}) \\
&\overset{M5,\,M3}{\Rightarrow} 0 < 1\cdot b^{-1} < (b\cdot b^{-1})\cdot a^{-1} \\
&\overset{M4,\,M5}{\Rightarrow} 0 < b^{-1} < 1\cdot a^{-1} \\
&\overset{M4}{\Rightarrow} 0 < b^{-1} < a^{-1}. \qquad \square
\end{aligned}$$

<u>Def</u> Let $(F,+,\cdot,<)$ be an ordered field, and let $A\subseteq F$ be nonempty. We say:

① A is <u>bounded above</u> if $\exists\, M\in F$ s.t. $a\leq M$ $\forall a\in A$. M is called an <u>upper bound</u> for A. If also $M\in A$, then M is called the <u>maximum</u> of A.

② A is <u>bounded below</u> if $\exists\, m\in F$ s.t. $a\geq m$ $\forall a\in A$. Then m is called a <u>lower bound</u> for A. If also $m\in A$, then m is called the <u>minimum</u> of A.

③ A is <u>bounded</u> if it is both bounded above

and bounded below.

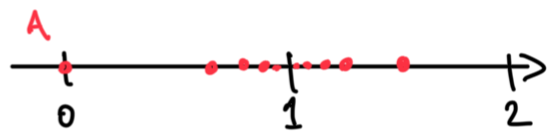
<u>Ex</u> $A = \{ 1 + \frac{(-1)^n}{n} : n \in \mathbb{N} \}$ is bounded.

3 is an upper bound for A.

$\frac{3}{2}$ is the maximum of A.

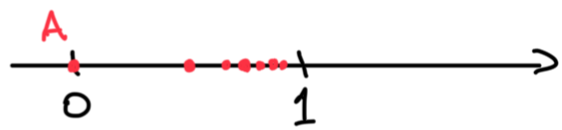
0 is a lower bound for A.

0 is the minimum of A.



<u>Ex</u> $A = \{ 1 - \frac{1}{n} : n \in \mathbb{N} \}$.

1 is an upper bound for A.



Claim: A does not have a maximum. Suppose not: $\exists M \in A$ s.t. $a \leq M$ $\forall a \in A$. As $M \in A$, $\exists n \in \mathbb{N}$ s.t. $M = 1 - \frac{1}{n}$. Then $1 - \frac{1}{n+1} \in A$, and

$$0 < n < n+1 \quad \Rightarrow \quad 0 < \frac{1}{n+1} < \frac{1}{n}$$

$$\Rightarrow \quad \underbrace{1 - \frac{1}{n+1}}_{\in A} > 1 - \frac{1}{n} = M$$

a contradiction.

<u>Rmk</u> • Some sets do not have a maximum.

• The maximum and minimum of a set are unique, if they exist. E.g. if $M, M' \in A$ are both maxima,

$$\left.\begin{array}{l} a \leq M \ \forall a \in A \quad \Rightarrow \quad M' \leq M \\ a \leq M' \ \forall a \in A \quad \Rightarrow \quad M \leq M' \end{array}\right\} \Rightarrow M = M'$$

<u>Def</u> Let $(F, +, \cdot, <)$ be an ordered field and let $A \subseteq F$ be nonempty. We say:

① $L \in F$ is the <u>least upper bound</u> (or <u>supremum</u>) of A if it satisfies:

- L is an upper bound for A, and
- if M is an upper bound for A then $L \leq M$.

When this is true, we write $L = \sup A$.

② $l \in F$ is the <u>greatest lower bound</u> (or <u>infimum</u>) of A if it satisfies:

- l is a lower bound for A, and
- if m is a lower bound for A then $l \geq m$.

When this is true, we write $l = \inf A$.

<u>Lem</u> The least upper bound of a set is unique, if it exists.

<u>Pf</u>: Suppose $\emptyset \neq A \subseteq F$ and L, M are least upper bounds for A. Then

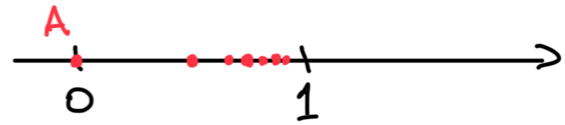
$$\left.\begin{array}{r} L \text{ least upper bound} \Rightarrow L \text{ upper bound} \\ M \text{ least upper bound} \end{array}\right\} \Rightarrow M \leq L$$

Similarly,

$$\left.\begin{array}{r} M \text{ least upper bound} \Rightarrow M \text{ upper bound} \\ L \text{ least upper bound} \end{array}\right\} \Rightarrow L \leq M.$$

So $L = M$. $\square$

<u>Ex</u> $A = \{1 - \frac{1}{n} : n \in \mathbb{N}\} \subseteq \mathbb{Q}$.



Claim: 1 is the least upper bound of A.

- 1 is an upper bound for A: $1 - \frac{1}{n} \leq 1 \quad \forall n \in \mathbb{N}$
- If $M \in \mathbb{Q}$ is an upper bound then $M \geq 1$:

Suppose not. Then

$$M = \frac{a}{b} < 1 \implies 0 < 1 - \frac{a}{b} = \frac{b-a}{b}, \quad b \in \mathbb{N},\ a \in \mathbb{Z}$$

$$\implies b - a \geq 1$$

$$\implies \frac{1}{b+1} < \frac{1}{b} \leq \frac{b-a}{b}$$

$$\implies 1 - \frac{1}{b+1} > 1 - \frac{b-a}{b} = \frac{a}{b} = M$$

So M is not an upper bound for A.

<u>Rmk</u> Sets that don't have a maximum may still have a least upper bound.